Junru (Andrea) Ai

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EDUCATION

Northwestern University **Evanston, IL**
M.S. in Technology, People, and Policy (STEM) GPA: 3.93/4.0 Sep 2025 – Sep 2026

- **Relevant coursework:** Technology in Context; Technology, Emotion, and Social Relationships; Technology, Psychology, and Education Across Cultures; Navigating AI and Data Governance; Data, Ethics, and Policy
- **Academic Activities:** Delegate, International Association for Safe and Ethical AI (IASEAI '26), Paris, Feb 2026

Wuhan University of Technology **Wuhan, China**
B.Econ. in Finance GPA: 3.22/5.0 Sep 2021 – Jun 2025

- **Relevant coursework:** Econometrics; Probability Theory and Mathematical Statistics; Statistics (SPSS); Linear Algebra; Advanced Mathematics (Calculus); Python Programming; Game Theory

RESEARCH INTERESTS

(1) Human-Computer Interaction; (2) Human-AI Interaction; (3) Social Computing; (4) Trust and Safety
Focus: human relationships with AI, digital well-being, and safety-by-design in interactive systems.

RESEARCH EXPERIENCE

Culture, Brain, and Human Development Lab & Technology and Human Development Lab Evanston, IL
Research Assistant, Northwestern University (PI: Prof. Yang Qu) *Mar 2026 – Aug 2026*

- Authored a literature review on how AI-driven products and short-form video influencers shape adolescents' online experiences and well-being, informing the labs' research on adolescent digital well-being.
- Led full-text retrieval for a three-RA systematic review of AI and technology in K-12 education, securing 251 of 265 records (95%); audited DOI metadata and built a Python workflow to deliver an analysis-ready corpus.
- Pilot-tested in-scanner attention paradigms for an adolescent fMRI study after MRI safety training, evaluating task usability and session workflow to flag protocol issues before data collection.
- Contributed to parallel parent- and child-report survey instruments on family relationships and adolescent development.

The Brookings Institution Washington, D.C.
Capstone Research Fellow (Supervisor: Dr. Nicol Turner Lee) *Dec 2025 – Jun 2026*

- Restructured and authored the talent pillar of a U.S.–EU–China comparison of AI industrial policy, analyzing immigration, workforce, and retention policy through Chinese- and English-language sources.
- Co-authored two policy briefs (AI talent retention; AI model export controls) and a blog post, *Beyond the Chip Race*, retained as internal research; presented at a spring symposium and the MTePP Showcase.

PROFESSIONAL EXPERIENCE

StepFun (Foundation Model Developer) Beijing, China
AI Product Intern *Jan 2024 – Mar 2024*

- Benchmarked four content-moderation vendors on 1M multimodal test samples (text, images, and AI-generated avatars; about 30% sexual content), measuring true-positive and false-positive rates; the team adopted my recommendation.
- Analyzed sexually explicit user–AI chats to identify evasion patterns and classification ambiguities, helping refine a 15-category, three-tier sexual-content taxonomy and its prohibited-term lexicon.
- Built scenario-based safety evaluation suites and an edge-case error-log database; diagnosed false positives on AI-generated avatars and benchmarked safety practices across Chinese LLM products.

Guangdong Genius Technology (BBK Electronics group; children's smart devices) Dongguan, China
Content Safety / Product Management Intern *Jun 2024 – Nov 2024*

- Analyzed 422 user complaints on content-safety risks for child users and screened the top 1,000 books on BBK learning tablets through automated and manual review, setting up recurring checks on third-party apps and links.
- Scraped 1,040 comments from 230 social media accounts with Python and analyzed how families evaluate children's learning devices, informing product iteration.

MytePro Shenzhen, China
Data Governance Intern *Dec 2024 – Apr 2025*

- Developed Python-based data cleaning and validation workflows to process 30,000+ real estate project records, performing data standardization, duplicate detection, and consistency checks to support structured data integration.
- Applied R and SPSS for exploratory statistical analysis of project-level data, using descriptive statistics and comparative analysis to investigate variation across project categories and support data-driven reporting.

Tencent Shenzhen, China
Product Data Analyst Intern *May 2025 – Aug 2025*

- Engineered automated data pipelines integrating 1M+ order and payment records using Python and SQL, with multi-table integration, cleaning, deduplication, and scheduled updates; reduced manual reporting by 15+ hours weekly.
- Built data-driven analytics interfaces in Power BI and Tableau to examine user behavior and conversion funnels, enabling systematic monitoring of product engagement and operational performance.

SELECTED RESEARCH PAPERS AND MANUSCRIPTS

Erotic conversations with AI as emotional coping. Unpublished manuscript, Technology, Emotion, and Social Relationships (Prof. Claudia Haase), Northwestern University, 2026.

- Designed a theory-informed semi-structured interview protocol and conducted in-depth interviews with AI companion users on how they use AI to navigate stigmatized emotional and intimate experiences.
- Integrated evolutionary-functionalist, social constructionist, and psychoanalytic perspectives, identifying a recurring cycle of loneliness, short-term emotional relief, and rebound shame.
- Translated findings into six safety-by-design principles for AI companions, including post-interaction emotional aftercare and non-coercive attachment cues to reduce emotional dependency.

The psychology of influencers and fans: A scoping review of parasocial bonds, celebrity worship, and fan identity in adolescent development. Unpublished manuscript, Technology, Psychology, and Education Across Cultures (Prof. Yang Qu), Northwestern University, 2026.

- Conducted a scoping review of interdisciplinary research on parasocial relationships, celebrity worship, and fan identity in adolescence, following the Arksey and O'Malley framework and PRISMA-ScR.
- Developed a five-cluster conceptual taxonomy synthesizing fragmented findings across developmental psychology, media studies, and adolescent social development.
- Identified three research gaps: developmental trajectories, culture as an explanatory mechanism, and the transition from social belonging to psychosocial risk.

Delivery riders and algorithmic management: A China-U.S. comparative qualitative study design. Research proposal (with J. Han, L. Li, Y. Zhang, and Y. Yang), Technology in Context (Prof. Reed Stevens and Vien Nguyen), Northwestern University, 2026.

- Co-developed a China-U.S. comparative qualitative design examining how algorithmic management shapes food-delivery workers' decision-making and interactions with platform systems.
- Designed a multi-method strategy combining semi-structured interviews, ethnographic fieldwork, ride-along observations, and app walkthroughs, grounded in distributed cognition.
- Developed a coding framework separating workers' interpretations of algorithmic directives from their behavioral responses, enabling structured cross-context comparison.

WeChat across two interviews: A comparative interview analysis. Course project, Technology in Context (Prof. Reed Stevens and Vien Nguyen), Northwestern University, 2026.

- Independently conducted, transcribed, and translated two semi-structured Mandarin interviews, building a bilingual qualitative dataset on users' experiences with digital platform ecosystems.
- Applied infrastructural invisibility and context collapse in a cross-case analysis, linking individual narratives to broader patterns of platform dependence and mediated social interaction.

How a self-checkout system processes a grocery purchase: A distributed cognition field study. Course project, Technology in Context (Prof. Reed Stevens and Vien Nguyen), Northwestern University, 2026.

- Conducted a distributed cognition field study at a supermarket self-checkout, using observation and photo documentation to trace how a purchase moves across people, devices, and displays; proposed interface improvements.

SKILLS

- **Research methods:** semi-structured interviewing, qualitative coding and thematic analysis, field observation, survey design, usability pilot testing, scoping and systematic literature review, fMRI safety training
- **Tools:** Python, R, NVivo, Figma, Qualtrics, SQL, SPSS, Tableau, Power BI, Otter.ai, Zotero
- **Languages:** Mandarin (Native), English (Advanced)